

Full Stack Web Application Homework – End of May 2026

Objective: *Develop a fully functional full stack web application that demonstrates modern web development practices, including responsive UI design, dynamic front-end behavior, robust server-side logic, data validation, and persistent data storage. The application must be delivered in a GitHub repository and must be runnable on a personal computer. After cloning the repository, the application should run immediately or with minimal setup as described in the README.*

1. Front-End Requirements

- Build the user interface using HTML5, CSS3, and preferably the Bootstrap framework for responsive and consistent styling.
- Implement dynamic client-side behavior using JavaScript.
- Provide form-based user interactions for creating, editing, and managing data.
- Implement client-side validation to ensure that user input meets the required constraints before submission.
- Ensure the UI is intuitive, accessible, and mobile-friendly.

2. Back-End Requirements

- Implement the server-side application using a technology stack of your choice, preferably Node.js/Express.
- Provide API endpoints to support CRUD operations.
- Implement server-side validation to ensure data integrity and security.
- Handle errors gracefully and return meaningful HTTP status codes and messages.
- The application must run on a local personal computer without requiring cloud infrastructure.

3. Data Persistence

- Use a persistent storage solution such as local file storage or a lightweight database solution.
- Define a clear data model with appropriate relationships and constraints.
- Ensure correct implementation of data creation, retrieval, updating, and deletion.

4. Application Features

Your application must include at least:

- A user-facing interface for interacting with stored data.
- A form-based workflow (e.g., adding new items, editing existing ones).
- A dynamic list or table view that reflects stored data.
- Feedback mechanisms such as success/error messages and validation hints.

5. Installation & Execution Requirements

- The entire project must be uploaded to GitHub in a complete, runnable form.
- The repository must be clone-ready: after cloning, the application should run immediately or with minimal setup.
- **If you cannot ensure direct clone-and-run functionality**, the README must contain a precise, step-by-step description of:
 - installation,
 - configuration (including environment variables, database setup, etc.),
 - and execution on a personal computer.
- The README must also include:
 - Project overview
 - System requirements
 - Troubleshooting notes
 - Any additional configuration details needed for local execution

6. Code Quality & Documentation

- Use a clear and logical project structure.
- Write clean, maintainable code following best practices.
- Include comments where necessary.
- Provide API documentation if applicable.

7. EXAMPLES

- Mini Webshop
- Pizza Ordering System
- Event Ticket Booking System
- Inventory Management System
- Appointment Booking System (Hair Salon, Dentist, etc.)
- Hotel Room Booking System
- Issue Tracker (Bug Tracker)
- Incident Management System (Call Center)
- Subscription Management System
- Car Rental Management
- Online Pharmacy
- Logistics Package Tracking